

## Requirement Analysis

### Technology Stack

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 28 June 2026                                                |
| Team ID       |                                                             |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 2 Marks                                                     |

**Table-1: Components & Technologies Used**

| S.No | Component            | Description                                                | Technology                              |
|------|----------------------|------------------------------------------------------------|-----------------------------------------|
| 1.   | Programming Language | Core language for data processing and ML model development | Python 3.8+                             |
| 2.   | Data Handling        | Loading, cleaning, and analyzing the crop dataset          | Pandas, NumPy                           |
| 3.   | Machine Learning     | Training the classification model for crop prediction      | scikit-learn (Random Forest Classifier) |
| 4.   | Model Storage        | Saving the trained model for reuse without retraining      | Pickle (model.pkl)                      |
| 5.   | Frontend / Web App   | Building the user input form and displaying results        | Streamlit                               |
| 6.   | Visualization        | Exploratory data analysis charts and correlation heatmaps  | Matplotlib, Seaborn                     |
| 7.   | Version Control      | Source code management and project submission              | Git & GitHub                            |
| 8.   | Deployment           | Hosting the live web application publicly                  | Streamlit Community Cloud               |

**Table-2: Application Characteristics**

| S.No | Characteristic         | Description                                  | Technology                      |
|------|------------------------|----------------------------------------------|---------------------------------|
| 1.   | Open-Source Frameworks | All frameworks used are free and open-source | scikit-learn, Pandas, Streamlit |

|    |              |                                                                 |                              |
|----|--------------|-----------------------------------------------------------------|------------------------------|
| 2. | Security     | Input validation to prevent application crashes on invalid data | Exception handling in Python |
| 3. | Scalability  | Can handle multiple concurrent users via the deployed web app   | Streamlit Cloud auto-scaling |
| 4. | Availability | Publicly accessible 24/7 once deployed                          | Streamlit Community Cloud    |